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CSC:345-02
Project 3: Virtual Memory Manager

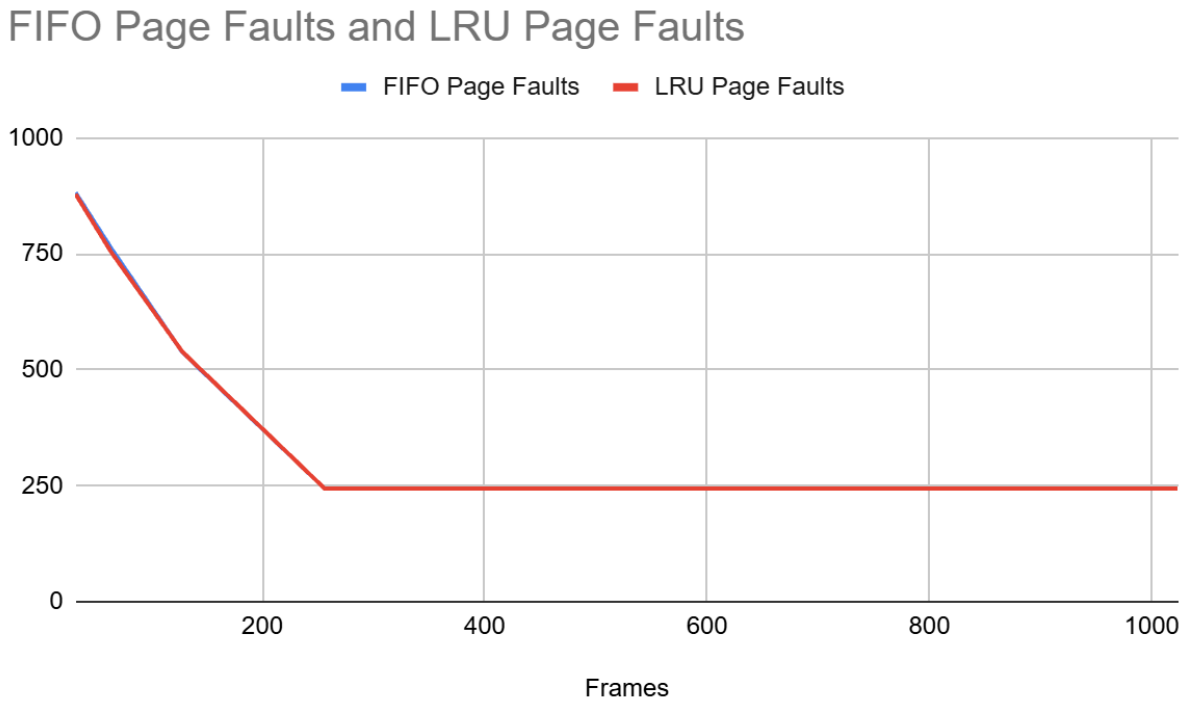
Requirements implemented:

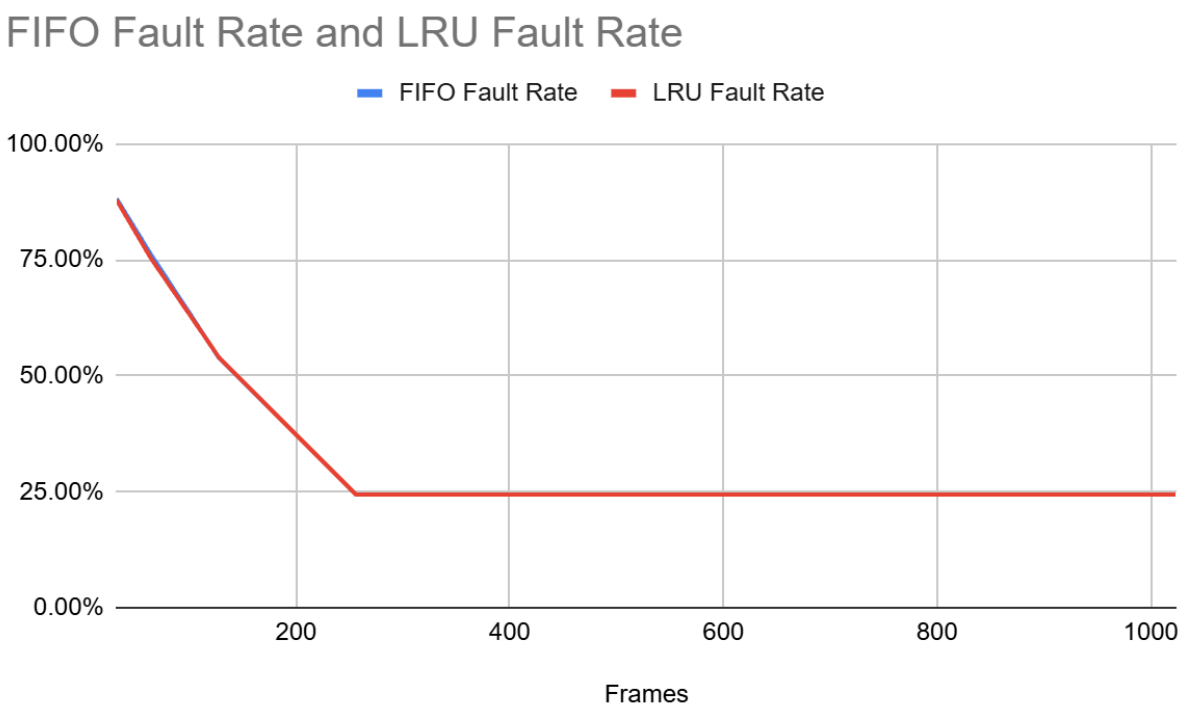
First, each given input address is separated into 8 bits for the local addresses and offset. The program then looks for the page in the TLB. On a TLB miss, it checks the page table. If the page is absent, a page fault occurs. On a page fault, the 256-byte page is read from BACKING_STORE.bin using fseek() and fread(). Once this is complete, or after the page was read if there was not a page fault, the physical address is computed (frame number x 256 + offset) and the byte at that position is retrieved. Each program records the logical address, translated physical address, and byte value of each input to out1.txt, out2.txt, and out3.txt, respectively to each program.

main1.c is demand paging without page replacement. It uses a 16-entry FIFO TLB, and can hold all 256 virtual pages at once. The program does not need to perform any evictions.

main2.c implements FIFO page replacement. Using a command-line argument, the program inputs the number of frames. A circular FIFO victim pointer selects the frame with the earliest loaded page. That page is evicted and invalidated, and removed from the TLB before a new page is loaded. TLB insertion is also done in FIFO order.

main3.c implements LRU page replacement. It also takes a frame count and uses a FIFO TLB. However, for replacing frames, it gives each one a last-used timestamp. The timestamp is updated on every address reference, including TLB hits. Oldest pages are evicted first instead of circularly.





Frames	FIFO Page Faults	FIFO Fault Rate	LRU Page Faults	LRU Fault Rate
32	882	88.20%	879	87.90%
64	762	76.20%	754	75.40%
128	538	53.80%	539	53.90%
256	244	24.40%	244	24.40%
512	244	24.40%	244	24.40%
768	244	24.40%	244	24.40%
1024	244	24.40%	244	24.40%

Analysis:
Both algorithms performances increased greatly with more frames. The size of memory is much more important for performance than the paging algorithm chosen. LRU preforms better with 32 and 64 frames, while FIFO preforms better with128 frames. The actual differences between the two are very close, and are not visible on the graphs. At 256 frames and above, both algorithms have the exact same number of page faults. This is because the given dataset only has 244 distinct pages. Adding frames beyond 256 therefore does not improve performances. TLB results are also low.